仿真环境下创建地图

# 1) 启动 roscore

gnome-terminal --title="roscore" -- bash -c "roscore"

# 等待 roscore 就绪

until rostopic list >/dev/null 2>&1; do sleep 0.2 done

# 2) 启动 Gazebo + robot

gnome-terminal --title="gazebo_start" -- bash -c "source ~/Gitkraken/ros1_slam/devel/setup.bash; roslaunch tri_steer_gazebo carto_test.launch"

# 等待 Gazebo 节点起来（/gazebo/model_states 出现再继续）

until rostopic list 2>/dev/null | grep -q "/gazebo/model_states"; do sleep 0.5 done

# 3) 启动 teleop + robot

gnome-terminal -t "teleop_start" -- bash -lc "source ~/Gitkraken/ros1_slam/devel/setup.bash; roslaunch tri_steer_keyboard keyboard_tri_steer.launch"

# 4) 雷达点云反转成水平车体下的point-cloud2

gnome-terminal -t "laser to cloud" -- bash -lc "source ~/Gitkraken/ros1_slam/devel/setup.bash; roslaunch read_laser_data laser_to_cloud.launch; exec bash"

# 5）启动cartographer建图节点

gnome-terminal -t "carto gmapping" -- bash -c "cd /home/action/Cartographer/Cartographer_Locatization;source install_isolated/setup.bash;roslaunch cartographer_ros my_robot_2d.launch;exec bash"

保存创建的地图

# 6）保存地图

cd /home/action/Cartographer/Cartographer_Locatization source install_isolated/setup.bash rosservice call /write_state "{filename: '/home/action/carto_map.pbstream'}"

# 7）转换 pbstream → 栅格地图

rosrun cartographer_ros cartographer_pbstream_to_ros_map -pbstream_filename=/home/action/carto_map.pbstream -map_filestem=/home/action/carto_map -resolution = 0.1

启动仿真环境下的导航

source ~/.bashrc set -e

# 1) 清理旧进程（允许失败）

rosnode kill -a 2>/dev/null || true killall -9 gzserver gzclient 2>/dev/null || true killall -9 roscore rosmaster 2>/dev/null || true

yes | rosclean purge

sleep 2

# 2) 启动 roscore

#ros_term "roscore" "roscore" gnome-terminal --title="roscore" -- bash -c "roscore"

# 等待 roscore 就绪

until rostopic list >/dev/null 2>&1; do sleep 0.2 done

# 3) 启动 Gazebo + robot

#这里面需要将地图更改成构建好的地图 gnome-terminal --title="gazebo_start" -- bash -c "source ~/Gitkraken/ros1_slam/devel/setup.bash; roslaunch tri_steer_gazebo sim_gazebo_rviz.launch"

# 等待 Gazebo 节点起来（/gazebo/model_states 出现再继续）

until rostopic list 2>/dev/null | grep -q "/gazebo/model_states"; do sleep 0.5 done

# 4) 启动 teleop + robot

# gnome-terminal -t "teleop_start" -- bash -lc "source ~/Gitkraken/ros1_slam/devel/setup.bash; roslaunch tri_steer_keyboard keyboard_tri_steer.launch"

# 5）启动地图节点

gnome-terminal -t "map start" -- bash -lc "source ~/Gitkraken/ros1_slam/devel/setup.bash; roslaunch robot_costmap map_deal.launch;exec bash"

# 6) 运动规划节点

gnome-terminal -t "motion Plan start" -- bash -lc "source ~/Gitkraken/ros1_slam/devel/setup.bash; roslaunch robot_navigation motionPlan_sim.launch"

# 7) 雷达点云反转成水平车体下的point-cloud2

gnome-terminal -t "laser to cloud" -- bash -lc "source ~/Gitkraken/ros1_slam/devel/setup.bash; roslaunch read_laser_data laser_to_cloud.launch; exec bash"

# 8）局部规划PID算法**********

gnome-terminal -t "PID_control" -- bash -c "source ~/Gitkraken/ros1_slam/devel/setup.bash; roslaunch robot_pid_local_planner Omnidirectional_PID.launch; exec bash"

# 9) 启动并发布机器人真值里程计消息

gnome-terminal -t "truth Odom start" -- bash -lc "source ~/Gitkraken/ros1_slam/devel/setup.bash; roslaunch robot_locatization truth_odometry.launch; exec bash"

# 10）订阅控制消息

gnome-terminal -t "control start" -- bash -lc "source ~/Gitkraken/ros1_slam/devel/setup.bash; roslaunch tri_steer_drive bringup.launch use_teleop:=true; exec bash"

仿真环境下的导航重定位

常用命令

#查看tf树 rosrun rqt_tf_tree rqt_tf_tree

产看 tf位姿 变化 rosrun tf tf_echo map base_footprint